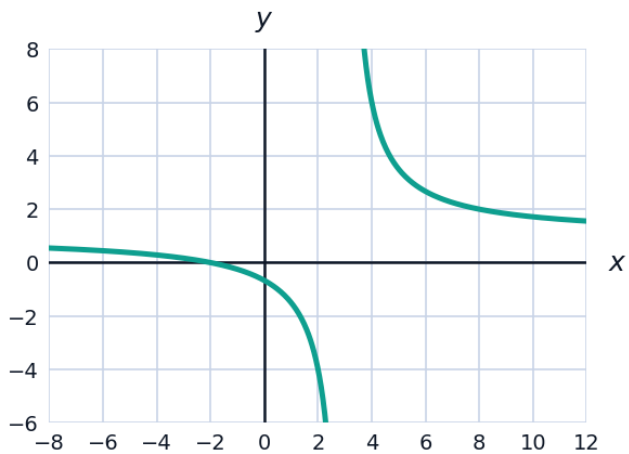
# Graphing Rational Functions

## Reading graphs

**1** The graph of $f(x) = \frac{x+2}{x-3}$ is shown. From the graph and the formula, state the asymptotes and intercepts.




**2** The function $g(x) = \frac{1}{x-2} + 3$ is a transformation of $y = \frac{1}{x}$.

**a** Describe the transformation.

**b** State the asymptotes of $g$.

**c** State the domain and range.

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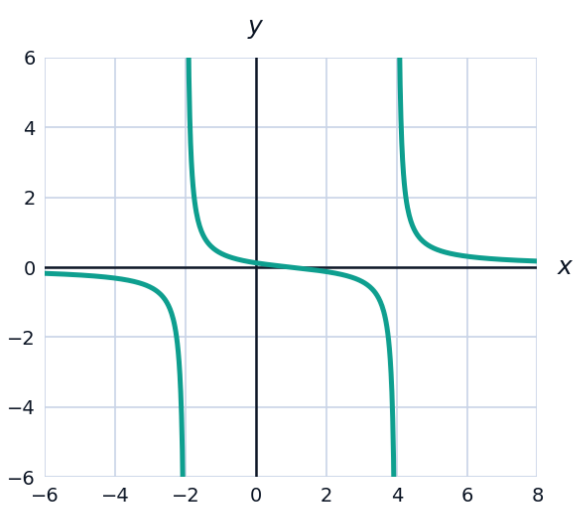
## Sign analysis

**3** Let $f(x) = \frac{x-1}{(x+2)(x-4)}$. Using a sign chart with critical values $x = -2, 1, 4$, determine the intervals on which $f(x) > 0$.

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4 Solve the rational inequality $\frac{x+3}{x-2} \leq 0$.

5 Sketch-planning: for $h(x) = \frac{2x}{x^2-1}$, state the asymptotes and the symmetry of the graph.

## Sketching a complete graph

6 For $f(x) = \frac{x^2-1}{x^2-4}$, find all intercepts and asymptotes, then describe the graph's behaviour near each vertical asymptote.

7 Sketch-planning: the function $g(x) = \frac{3}{x-1} - 2$ has a vertical asymptote at $x = 1$ and horizontal asymptote $y = -2$. Find its $x$- and $y$-intercepts.

## More rational inequalities

8 Solve $\frac{(x-2)(x+1)}{x-4} \geq 0$.

9 Solve $\frac{2}{x+3} > 1$.

## Building a rational function from a description

10 Construct a rational function with vertical asymptote $x = 2$, horizontal asymptote $y = 1$, and a hole at $x = -3$.

## Graphing with a slant asymptote

11 For $f(x) = \frac{x^2-4}{x-1}$, find the slant asymptote and the vertical asymptote, and use them together with intercepts to describe the overall shape of the graph.

## Full graph analysis: putting it all together

12 For $f(x) = \frac{2x^2-2}{x^2-9}$, find all intercepts, all asymptotes, and describe the symmetry of the graph (even, odd, or neither).

## A compound rational inequality

13 Solve $\frac{x^2-1}{x-2} \leq 0$.

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## Graphing near a hole

**14** For $f(x) = \frac{x^2 - 9}{x - 3}$, describe how the graph looks immediately around $x = 3$, and find the coordinates of the hole.

**15** Solve $\frac{3}{x+1} \geq \frac{2}{x-1}$.

**16** A rectangular garden's cost per square meter to fence is modeled by $C(x) = \frac{200}{x} + 8x$, where $x$ is the garden's width in meters. Find the vertical asymptote and horizontal/slant asymptote of this function, and interpret what the vertical asymptote means physically (why $x = 0$ isn't a realistic input).

**17** State the domain of $f(x) = \frac{\sqrt{x}}{x - 4}$.

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